GUI Programming in Python (Tkinter) – Notes

📌 1. Introduction to GUI

🔹 What is GUI?

GUI (Graphical User Interface) allows users to interact with software using:

- Buttons
- Icons
- Text fields

👉 Instead of typing commands, users **click and interact visually**

🔹 Examples of GUI Applications

- Calculator
- Notepad
- Web browsers
- Mobile apps

📌 2. GUI vs CLI

Feature	GUI	CLI
Interaction	Visual (buttons, icons)	Text commands
Ease of Use	Easy for beginners	Requires knowledge
Examples	Calculator, Chrome	Terminal, CMD

📌 3. What is Tkinter?

🔹 Definition

Tkinter is Python’s built-in library for creating GUI applications.

👉 It is:

- Simple
- Beginner-friendly
- No extra installation required

🔹 Import Tkinter

```
importtkinterastk
```

📌 4. Basic Window Setup

🔹 Create Main Window

```
importtkinterastk

root=tk.Tk()
root.mainloop()
```

🔹 Important Components

Component	Description
`Tk()`	Creates main window
`mainloop()`	Runs the application

Component	Description
`title()`	Sets window title
`geometry()`	Sets window size

🔹 Example

```
root.title("My App")
root.geometry("400x300")
```

📌 5. Key Widgets in Tkinter

Widgets are elements placed on GUI.

🔹 1. Label

Displays text

```
tk.Label(root,text="Hello").pack()
```

🔹 2. Entry

Takes user input

```
entry=tk.Entry(root)
entry.pack()
```

🔹 3. Button

Triggers actions

```
tk.Button(root,text="Click Me").pack()
```

🔹 4. Canvas

Used for drawing shapes/images

📌 6. Layout Management

Controls how widgets are arranged.

🔹 1. Pack Layout

- Simple
- Auto-arranges widgets

```
widget.pack()
```

🔹 2. Grid Layout

- Table-like structure

```
widget.grid(row=0,column=1)
```

🔹 3. Place Layout

- Exact positioning

```
widget.place(x=50,y=100)
```

📌 7. Event Handling

🔹 What is Event?

An **event** is an action like:

- Button click
- Key press

🔹 Example: Button Click

```
defclick():
print("Button clicked")
```

```
tk.Button(root,text="Click",command=click).pack()
```

📌 8. Mini Project: Login Form

🔹 Features:

- Username input
- Password input
- Login button

🔹 Example Code

```
importtkinterastk

deflogin():
username=user.get()
password=pwd.get()
print(username,password)

root=tk.Tk()

tk.Label(root,text="Username").pack()
user=tk.Entry(root)
user.pack()

tk.Label(root,text="Password").pack()
pwd=tk.Entry(root,show="*")
pwd.pack()

tk.Button(root,text="Login",command=login).pack()

root.mainloop()
```

📌 9. Advanced Project: Finance Tracker

🔹 Features:

- Add income/expenses
- Store data
- Show reports
- User-friendly interface

📌 10. Real-World Applications

✅ Inventory System

✅ File Organizer

✅ Billing Software

✅ Login Systems

👉 Tkinter is used to build **desktop applications**

📌 11. Best Practices

- Keep UI simple
- Use proper layout
- Label everything clearly
- Maintain consistent design

📌 12. Common Mistakes ⚠️

- Spelling mistakes in `geometry()`
- Forgetting `mainloop()`
- Wrong widget placement
- Not handling events properly

📌 13. Final Understanding

If students understand:

- Window creation
- Widgets
- Layout
- Events

👉 They can build:

- Desktop apps
- Forms
- Tools

📌 14. Key Takeaway

> GUI makes programs user-friendly
>
> Tkinter makes GUI easy in Python